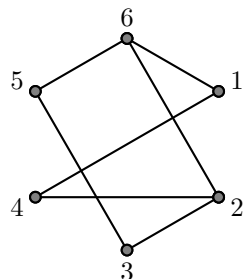


Final Exam

1 Answer the following questions about the graph below.

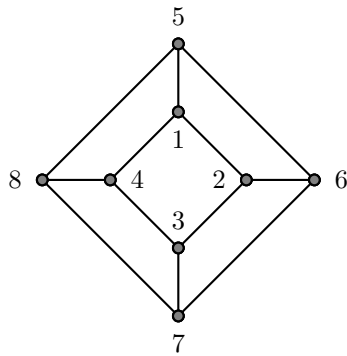


- a) What is an Euler circuit?
- b) Does the graph have an Euler circuit? How do you know?
- c) Does the graph have an Euler path that is not a circuit? How do you know?
- d) What is a Hamilton circuit?
- e) Does the graph have a Hamilton circuit? How do you know?

2 Draw a simple graph with 7 vertices, each of which has degree 4.

3 Explain why it is impossible to draw a simple graph with 7 vertices, each of degree 3.

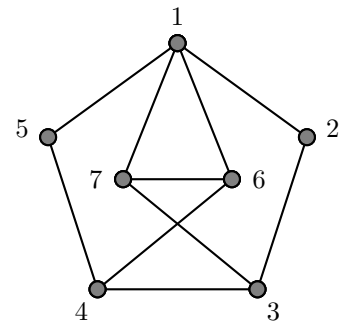
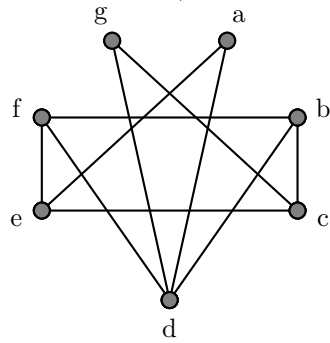
4 Is the graph below bipartite? Explain.



5 Give an example of a graph that is not a planar graph and explain how you know it is not planar. **Do not use Kuratowski's Theorem.**

6 How many isomorphisms are there between the following two graphs? **Explain.**

(If you do not know exactly how many there are, tell me as much as you know about how many there are. Examples: There is at least 1; there are at most 3, etc.)



7 COINS AND CARDS.

- a) If I flip a coin 5 times, what is the probability of getting the same result each of the five times?
- b) If I deal five cards from a standard 52 card deck, what is the probability that they will all be the same color.
(The deck has 26 red cards and 26 black cards.)

8

- a) Show how to compute $C(11, 5) = \binom{11}{5}$ without using a calculator.
- b) Write a counting problem for which $\binom{11}{5}$ is the answer. [Keep it simple.]

9 ABC-123 LOTTERY. Consider the following lottery type game: A player selects (in order) three *distinct* letters from A to Z and three *distinct* digits from 0 to 9. (Examples: one lottery ticket would be AZD-924, another would be ADZ-492. On the other hand, ABA-012 is not a legal ticket because A is repeated and 123-ABC is not a legal ticket because the numbers come first.) Later, the lottery commission draws three distinct letters and three distinct digits at random and in order.

- If all the numbers and letters match in order, then the player wins \$1,000,000.
- If all the numbers and letters match but are not in the correct order, then the player wins \$100,000.
- If the all the letters match (not necessarily in order), but the numbers do not, then the player wins \$1,000;
- If the all the numbers match (not necessarily in order), but the letters do not, then the player wins a prize of \$10;
- In all other cases, the player wins nothing.

a) Fill in the following table. Show your work in the space below.

amount won	1,000,000	100,000	1000	10	0
probability					

b) Determine the **expected value** of the amount won from one lottery ticket. [This will be graded based on your answers to the questions above, even if they are incorrect, so be sure to show your work.]

c) How much must the lottery charge per play to avoid losing money at this game?

d) For which kind of prize should the lottery commission expect to pay out the most money? Explain.

10 Sam collects commemorative stamps. Among his favorites are Maria Goeppert Mayer stamps  and the new cowaphant stamps. 

On Sam's desk are four envelopes with stamps in them. The table below shows how many of each kind of stamp are in each envelope.

envelope	 Maria Goeppert Mayer	 cowaphant
A	1	4
B	3	7
C	10	10
D	18	2

Sam randomly selects an envelope (each envelope equally likely) and then randomly selects a stamp from within that envelope (each stamp in the envelope equally likely).

a) What is the probability that Sam chooses a cowaphant stamp?

b) If Sam chooses a cowaphant stamp, what is the probability that the envelope was envelope A?

11 Dice.

- a) If you roll three dice (each with six sides numbered 1, 2, 3, 4, 5, and 6), what is the probability that the three numbers rolled will all be the same?
- b) If you roll three dice (each with six sides numbered 1, 2, 3, 4, 5, and 6), what is the probability that the three numbers rolled will be distinct? (no repetitions)

12 What is the output of the following python code?

```
>>> import re as re
>>> re.findall(r'a+b*[cd]*', 'abcdabccddcbaabbccddccbbaa')
```

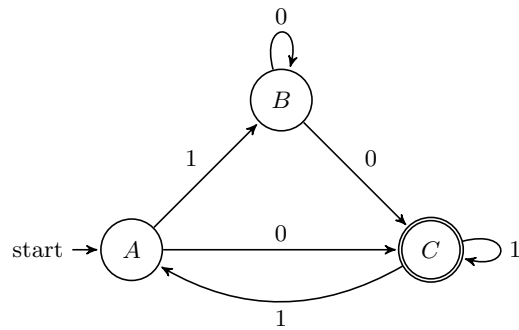
13 Let L be the language consisting of all binary strings that end with at least two 0's followed by at least one 1. Examples: $10011 \in L$, $0010 \notin L$.

a) Write a regular expression for L .

b) Construct a DFA that recognizes L .

14 Give an example of a language that is not regular, and explain how you know the language is not regular.

15 Below is the state diagram for an NFA N .



a) Which of the following strings are accepted by N ? (Simply write “accept” or “reject” after each.)

- | | |
|--------|---------|
| • 101 | • 1001 |
| • 0011 | • 0101 |
| • 0100 | • 01100 |

b) Design a DFA that accepts the same language as N using a general method that works for all NFAs.

c) Construct a regular grammar for this language using a general method that works for all NFAs.

16 Below is the instruction set for a Turing machine M with start state s_0 and final state (i.e., accepting state) s_4 . In each row, the columns are in the order state, symbol, state, symbol, direction.

s_0	0	s_1	X	R
s_0	1	s_1	1	R
s_1	0	s_1	0	R
s_1	1	s_2	1	R
s_2	0	s_1	0	R
s_2	1	s_3	Y	L
s_3	0	s_4	0	R
s_3	1	s_4	Y	R

Fill out the table below:

input string	Does M accept? (Y or N)	tape contents upon halting (or DNH for does not halt)	last state (or DNH for does not halt)
0011			
1101			
0101			

This space available for extra work, cowaphant art, etc.

9

```

vals <- c(1000000, 100000, 1000, 10, 0)
plays <- (26*25*24*10*9*8); plays

## [1] 11232000

probs <- 1/plays * c(
  1, # perfect match (in order)
  3 * 2 * 1 * 3 * 2 * 1 - 1, # correct but in wrong order
  3 * 2 * 1 * (10 * 9 * 8 - 3 * 2 * 1), # letters correct, numbers not
  3 * 2 * 1 * (26 * 25 * 24 - 3 * 2 * 1) # numbers correct, letters not
)
probs <- c(probs, 1 - sum(probs)); probs

## [1] 8.903134e-08 3.116097e-06 3.814103e-04 8.330128e-03 9.912853e-01

probs * plays

## [1] 1 35 4284 93564 11134116

sum(probs * vals)

## [1] 0.8653526

probs * vals

## [1] 0.08903134 0.31160969 0.38141026 0.08330128 0.00000000

1/probs

## [1] 1.123200e+07 3.209143e+05 2.621849e+03 1.200462e+02 1.008791e+00

1 /
c(11232000,
  (choose(26, 3) - 1) * (choose(10, 3) - 1),
  (choose(26, 3) - 1) * choose(10, 3) / (choose(10, 3) - 1),
  (choose(10, 3) - 1) * choose(26, 3) / (choose(26, 3) - 1),
  1) /
probs

## [1] 1.000000 1.037614 1.000385 1.008403 1.008791

```

10

```

sum(.25 * c(4/5, 7/10, 1/2, 1/10))
## [1] 0.525

sum(.25 * c(4/5, 7/10, 1/2, 1/10)) %>% fractions()

## Error in sum(0.25 * c(4/5, 7/10, 1/2, 1/10)) %>% fractions(): could not find function "%>%"

.25 * c(4/5, 7/10, 1/2, 1/10) / sum(.25 * c(4/5, 7/10, 1/2, 1/10))

## [1] 0.38095238 0.33333333 0.23809524 0.04761905

.25 * c(4/5, 7/10, 1/2, 1/10) / sum(.25 * c(4/5, 7/10, 1/2, 1/10)) %>% fractions()

## Error in sum(0.25 * c(4/5, 7/10, 1/2, 1/10)) %>% fractions(): could not find function "%>%"

```

12

```

>>> import re as re
>>> re.findall(r'a+b*[cd]*', 'abcdabcddcbaabbccddccbbaa')
['abcd', 'abcddc', 'aabbccddcc', 'aa']

```

15 $(01 \mid 10^*01^*1)^* 10^*01^*$